

Troubleshooting Methodology -- Guided Notes

Unit 4.1 | CompTIA Network+ N10-009 Obj. Core 2 5.1 | N10-008

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The CompTIA troubleshooting methodology has _____ steps. Step 1 is _____ the problem. Step 2 is _____ a theory of probable cause. When gathering information in Step 1, open-ended questions (e.g., 'What were you doing when this happened?') are preferred over closed questions because they _____.

Question 2

Step 3 is _____ the theory. If the theory is confirmed, proceed to Step 4. If the theory is _____, establish a new theory or escalate. Step 4 is establishing a/an _____ of action to resolve the problem and identifying the potential effects of the fix.

Question 3

Step 5 is _____ the solution or escalating as necessary, then _____ full system functionality and -- if applicable -- implementing preventive measures. Step 6 is _____ findings, actions, and outcomes, which creates a reference for future repairs and protects the technician legally.

Question 4

Step 7 is the final step: _____ the user. This includes explaining what caused the problem in non-technical terms, showing the user how to prevent the issue in the future, and confirming _____ with the resolution. This step is often skipped by junior technicians but is important for _____ and professional reputation.

Question 5

The distinction between a _____ and a _____ is critical in troubleshooting. The first is what the user observes or reports (e.g., 'the screen is black'). The second is the underlying technical reason (e.g., 'the GPU is not seated'). Treating a symptom without finding the _____ will result in the problem recurring.

Question 6

Escalation means passing the problem to a _____ skill level or different team when it is beyond the current technician's ability to resolve. Before escalating, the technician should document what has already been _____ and what the _____ confirmed or ruled out through testing.

Question 7

When should a technician consider whether a _____ change occurred recently? If a user says 'it just stopped working,' a technician should ask about any _____, _____, or _____ changes in the recent past, as these are the most common causes of sudden failures.

Question 8 -- Real World Application

REAL WORLD: A user calls and says, 'My computer doesn't work.' Using the seven-step troubleshooting methodology, describe the exact questions you would ask in Step 1 and how those questions lead to a testable theory in Step 2. Demonstrate the difference between asking open-ended and closed questions.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-1-troubleshooting-methodology/4-1-troubleshooting-methodology-answer-key.pdf
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